

APPROVED METHOD:
LABORATORY OR FIELD THERMAL
MEASUREMENTS OF FLUORESCENT LAMPS
AND BALLASTS IN LUMINAIRES
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-62-20(R2023)

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Publication of this document
has been approved by the IES.
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should be directed to IES.

**Prepared by the
IES Testing Procedures Committee**



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Foreword

This Approved Method, which is a revision of *IESNA LM-62-06, Laboratory or Field Measurements of Fluorescent Lamps and Ballasts in Luminaires*, is for luminaire designers and manufacturers to use in improving equipment performance. As an IES Approved Method, it is intended as an aid to the designer when measuring operating temperature of lamps and ballasts in luminaires under either laboratory or field conditions.

1.0 Introduction

1.1 Scope

This Approved Method covers only thermal measurement of fluorescent lamps and ballasts in luminaires. Its purpose is to aid luminaire designers in achieving optimum performance of these components in given applications. In addition to the general test procedures outlined in this document, lamp and ballast manufacturers' data sheets should always be consulted when possible. Manufacturers of these products often have technical guides available, detailing product-specific thermal test point locations and limits. Manufacturers should be contacted for these guides for the products used in the design.

1.2 Need for Thermal Testing

Fluorescent lamps are temperature sensitive light sources. For most fluorescent light sources, their light output, power consumption, and efficacy are influenced by the temperature of the tube at its coolest point (the "cold spot" or "cool spot"), which determines mercury vapor pressure. (See **Annex A**) Temperatures greater than optimum will cause lamp lumens and power consumption to decline nearly proportionally; however, at temperatures cooler than optimum, light output may decline rapidly while power consumption remains high, causing a precipitous reduction in efficacy. High efficiency lamp and ballast systems coupled with heat sinking metal louvers and/or heat removal luminaires can have a surprisingly adverse effect unless temperature factors are carefully considered.

Conversely, the critical temperature affecting ballast performance is the hottest part of the ballast. For typical electromagnetic ballasts, this is the temperature of the transformer core and coil. It is generally measured on that part of the ballast case closest to the coil. The location of the hot spot can be obtained from the ballast manufacturer or can be determined by carefully scanning the case with a temperature probe. The importance of ballast temperature rise is apparent from the industry recognized fact that a rise of 10 °C (18 °F) in coil temperature above 90 °C (194 °F) can reduce ballast coil life by 50 percent. Since 1968, all U.S. domestic indoor high power factor fluorescent ballasts have been equipped with built in Class P thermal protectors.^{1,2} These Class P protectors are designed to switch the ballast primary off when the maximum case temperature rises in excess of 100 °C (212 °F). The ballasts will restart the lamps in about 30 minutes when the case cools to about 90 °C (194 °F).

Electronic type ballasts will typically have a maximum case temperature less than 90 °C (194 °F). The manufacturer's temperature limit for an electronic ballast should be adhered to closely; otherwise, ballast life and performance could be greatly reduced.

2.0 Measuring Equipment

There are many types of instruments suitable for temperature measurements of fluorescent lamps and ballasts. It is recommended that the temperature sensing device have very low mass and thermal inertia so that it will not influence the temperature of the point being measured. One such device that is widely used is a 30-gauge iron constantan thermocouple (type J). This may be used with a digital meter capable of reading Celsius or Fahrenheit degrees with an accuracy of ± 0.5 °C (± 1 °F). All measurement equipment shall be calibrated annually by a recognized calibration laboratory, traceable to NIST (U.S. National Institute of Standards and Technology).

Since most luminaire tests involve several temperature points to be monitored, a rotating selector switch

will permit a large number of thermocouples to be monitored by a single meter. The selector switch shall be of the type designed for switching thermocouples; i.e., it shall have very low contact resistance and shall be designed not to introduce thermoelectric effects of its own. Automated, multiplexing-type data acquisition systems may also be used for this application, provided the appropriate signal conditioning is implemented.

Thermistors may also be used for temperature sensors, provided they exhibit stable characteristics and are of sufficiently small size and shape to allow attachment to the lamp or ballast surfaces. Caution should be observed if thermistors are used, because some of them are sensitive to light as well as to temperature. The use of thermistors also requires additional electronic circuitry to excite the thermistors, which are usually operated with a constant-current source. Thermistors have the advantage of a very high responsivity, and therefore high temperature measurement resolution and accuracy. Measurement accuracy can exceed 0.1 °C (0.2 °F).

Non-contact, infrared-based instruments are available for temperature measurement of ballast and lamp temperature points. These instruments are useful for quickly determining key measurement points for thermocouple measurement or for measuring difficult location points not easily measured by direct contact means. Non-contact instruments should not be relied on entirely for temperature measurement, since their accuracy is generally not as good as with direct-contact measurement with thermocouples. Direct-contact measurements with thermocouples is the preferred measurement method whenever possible.

3.0 Laboratory Test Conditions

3.1 Thermal Environment

Measurements shall be made in a draft free room with a maximum airflow of 0.08 m/s (15.8 ft/min). The ambient temperature shall be maintained at 25 °C $\pm$ 1 °C (77 °F $\pm$ 2 °F), unless specifically testing for operation in other ambient temperatures. Temperature should be measured within 1 m (3.3 ft) of the luminaire and shielded from direct radiation.

All reported temperatures (measured in a laboratory) should be corrected to reflect an ambient temperature of 25 °C (77 °F). For example, if the recorded temperature measurement is 60 °C and the ambient is 24 °C, then the final reported temperature would be 61 °C.

3.2 Thermal Stabilization

The period of time required for a lamp to achieve thermal stabilization will vary according to the amount and location of the mercury within the lamp. The lamp will be stable when most of the mercury has condensed at its coldest point. The ballast and the lamp(s) in a typical T8 or T12 enclosed recessed troffer or surface-mounted luminaire will usually require about six hours to reach thermal equilibrium; other lamp types, such as T5 linear, T12 high output, and compact fluorescent lamps, may require as long as 15 hours to become thermally stable. Temperature readings (on ballasts or lamps) are considered stabilized when three successive readings taken at 30 minute intervals are within 1 °C (1.8 °F) of one another and are not rising.

3.3 Lamps and Ballasts

Lamps should be selected that are typical of latest production and conform to the latest requirements of the American National Standards Institute.³⁻⁷ Where ANSI standards do not exist for a given lamp type, values published or stated by the lamp manufacturer should apply. Lamps should be aged in accordance with ANSI/IES LM-54-20.⁸ Ballasts chosen for the tests should be typical of current production.

3.4 Luminaires

Luminaires chosen for the tests should be typical of the latest production and should be mounted as closely as possible to their normal field conditions.

3.5 Line Voltage

The ballast(s) in the luminaire should be operated at their rated primary voltage and frequency ($\pm 0.5\%$). When ballasts are labeled for a range of primary voltages, the ballast should be operated at the intended application voltage. The applied voltage shall have a sinusoidal wave shape such that the root-mean-square (RMS) summation of the harmonic components will not exceed 3% of the fundamental.

3.6 Field Test Conditions

If lamp and/or ballast temperature measurements are to be made in other than controlled laboratory conditions, any variations from the conditions stated previously shall be noted in a descriptive test report. In addition, any ambient conditions that may affect the luminaire temperatures, such as plenum temperature or proximity of thermal insulating materials, shall be described in the report (see **Section 6.2**).

4.0 Location and Attachment of Temperature Sensor on Fluorescent Lamp

4.1 Location of Temperature Sensor on Linear Fluorescent Lamps

The temperature sensor should be located on the coolest portion of the tube. This is a critical spot that influences the electrical and photometric performance of the lamp.

As a general rule, in luminaires with T12 lamps the cool spot occurs in the bottom-center portion of the tube. Typically, on T8 lamps the cool spot may occur at the end of the glass tube. For all lamp types it is advisable to check with the lamp manufacturer to confirm lamp cool-spot location for a given lamp-ballast combination. A temperature of approximately 38 °C (100 °F) measured at the cool-spot location will usually provide optimum performance of T8 and T12 lamps. When using luminaires that are ventilated and/or have metal louvers in close proximity, it is advisable to allow the lamps to stabilize thermally, then to place the sensor in the spot where the mercury can be seen visually. This frequently occurs where the louvers come closest to the tube.

T12 lamps designed to operate at 1500 mA have special features to create cool spots to control mercury vapor pressure. These T12 types have heat-reflecting shields directly behind (base side) the electrodes. These end chambers then become the control or cold spots. The noncircular cross section of a 1500-mA lamp has two grooves in the center of the lamp deeper than all the

others. The rails along side these grooves are then farther from the arc and become cold spots.

T5 linear lamps are designed to produce optimum light output at a higher ambient temperature (35 °C [95 °F]) than T8 and T12 lamps. The cool-spot location for T5 lamps is located on the etched end of the lamp, behind the electrode. As used in this section, “etched end” refers to the location where the manufacturer places the lamp’s identifying information. A temperature of approximately 42 °C (108 °F) measured at the cool-spot location will usually provide optimum performance of T5 lamps. T5 lamps should be pre-burned for 15 hours in a vertical position with the cool-spot down to encourage mercury migration before testing. After pre-burning, the lamp(s) should be carefully moved (etched end down) in a vibration-free manner to the luminaire for final thermal testing, positioning the cool-spot location in the lowest allowable position in the luminaire.

Lamps with special mechanical devices to induce a cool spot require extra care. Such devices include metallic heat sinks, clips, or springs that come into contact with the lamp wall and are designed to conduct heat away from the lamp. The location of contact with the lamp will likely constitute the cool spot.

4.2 Location of Temperature Sensor on Nonlinear Fluorescent Lamps

Extra care should be taken when making thermal measurements of compact, U shaped, circular, or other nonlinear lamps. For these lamps, the cool-spot location and value can change dramatically depending upon operating position (base up, base down, or horizontal) and if metal clips are used to support the lamps and the luminaire. Many compact fluorescent lamps will typically have very specific cool-spot, base, and guide post measurement locations and temperature limits. The lamp manufacturer should be consulted for this information.

The light output of compact fluorescent lamps that contain amalgam (usually higher-wattage types) are more influenced by the ambient temperature around the lamp than any specific cool-spot temperature measurement point. In this case, a temperature sensor should be placed

in the lamp cavity, approximately 25 mm (1 in.) away from the lamp in the luminaire. In addition, this sensor should be shielded from direct radiation from the lamp. The lamp manufacturer should be consulted for the optimal ambient temperature range for these particular lamp types.

Unless experience indicates otherwise, all compact fluorescent lamps should be allowed to stabilize for 15 hours before initial temperature measurements are taken, to allow the mercury to completely stabilize within the lamp. Subsequent tests may require less stabilization time if lamps are protected against bumps or jolts during handling that could effect the distribution of mercury in the lamp.

4.3 Attachment to Fluorescent Lamp

When using thermocouples, it is important that the thermocouple junction be maintained in firm thermal contact with the tube. This is accomplished by cementing the junction to the glass using sodium silicate (water glass) with fuller's earth or by using a commercial two component thermocouple cement (e.g., Omega® CC High Temperature Cement). Good adhesion can also be attained with cyanoacrylate-based adhesives (e.g., Superglue®). Since fluorescent tubes are coated with silicone, it is advisable to clean this off with an appropriate solvent. Adhesion is sometimes enhanced by lightly sanding the glass prior to applying the cement; however, glass should not be abraded at stress points such as bends. This can cause immediate or latent glass failure. In addition, it is recommended that the thermocouple wire be held to the tube 25 to 50 mm (1 to 2 in.) away from the junction by a thin strip of white fiberglass tape to act as a strain relief. The thermocouple wire should be slightly bowed near the junction so that only the junction is in contact with the lamp at that point. For plastic-material lamp points, the thermocouple junction may be lightly melted into the plastic with a soldering iron.

5.0 Location and Attachment of Temperature Sensor on Ballast

5.1 Location on Ballast

The location of the temperature sensor should be at the hottest point on the ballast case. This can be found

by scanning the case with a temperature probe or by placing temperature sensors approximately every 25 mm (1 in.) along the centerline of the ballast case. Typically, manufacturers will indicate temperature measurement points directly on the ballast label (along with a maximum permissible temperature), or the specific location can be obtained from the ballast manufacturer.

5.2 Attachment to Ballast

The site on the ballast case where the thermocouple junction is to be placed is first prepared by scraping off the label and paint down to bare metal in an area about 12 mm (0.5 in.) square. The thermocouple junction is then welded or soldered (using rosin core solder) to the ballast case. Where the use of solder is not practical (as in the case of field measurements), adhesives may be used. For a plastic ballast case, the thermocouple junction may be lightly melted into the ballast case with a soldering iron.

Thermocouple leads should be strain-relieved by fiberglass tape approximately 50 to 75 mm (2 to 3 in.) from the thermocouple junction. If multiplexing data acquisition is used, it may be necessary to electrically insulate the thermocouple junction from the ballast case to avoid measurement errors due to undesirable grounding of the thermocouple junction. This can be accomplished with a thin layer of electrically insulating tape or by coating the thermocouple junction with a non-conducting medium. Care should be exercised to maintain good thermal contact between the thermocouple junction and the ballast case.

6.0 General Guidelines

6.1 Considerations

The careful measurement of fluorescent lamp and ballast thermal characteristics is important to luminaire designers. Designers need to be cognizant of variables that can dramatically affect their results. For example, metal louvers or reflectors in close proximity (less than 4 mm [0.16 in.]) to a lamp can move the cool spot and significantly lower the lamp's temperature. Vented luminaire design relying on natural convection can be used as a possible method of maintaining optimum

ambient luminaire temperature. Care shall be taken in employing this method by strategically placing the openings in order to avoid adversely affecting optical performance. An unusual property of high efficiency systems is that it is possible for lamps to operate too cool in some types of indoor luminaires. Prudent system designers need to be aware of this possibility and have accurate test data to give them proper design direction.

6.2 Test Report

The following parameters should be noted and reported in addition to thermal measurements taken of the lamps and ballasts.

Luminaire description:

- Luminaire type
- Manufacturer
- Lens or louver type
- Air handling: heat extract or static
- Lamp description
- Ballast description
- Any other noteworthy characteristics

Test conditions:

- Laboratory test or field test
- Ambient temperature
- Plenum temperature
- Luminaire backing, if any
- Ballast cavity temperature
- Lamp cavity temperature
- Line voltage
- Power consumed
- Time permitted for thermal stabilization
- Age of installation (field test only)
- Any other noteworthy comments

Annex A – Ambient Temperature Effects On Lamp Cool Spot

Note: The graphs in this Annex are provided for illustrative purposes only; lamp manufacturer should be consulted for specific data.

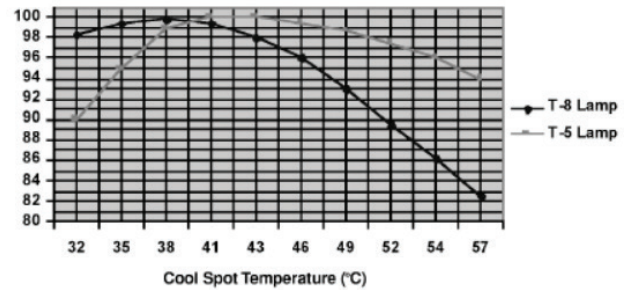


Figure A-1. Typical percent light output vs. cool spot temperature in T-8 and T-5 lamps.

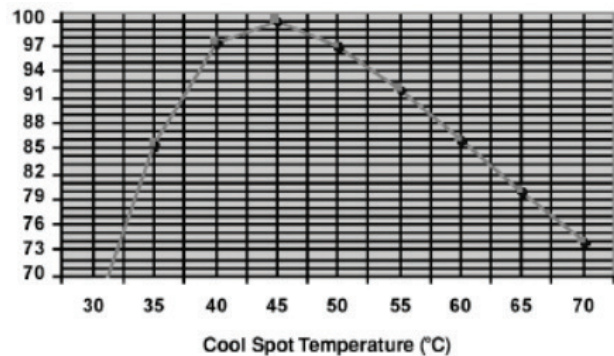


Figure A-2. Typical percent light output vs. cool-spot temperature in a compact fluorescent lamp (non-amalgam type, base up).

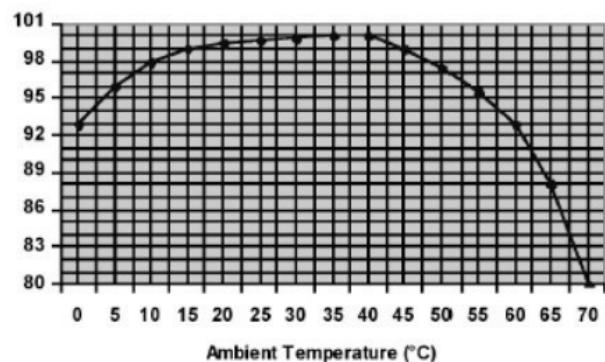


Figure A-3. Typical percent light output vs. ambient temperature in a compact fluorescent lamp (amalgam type, base up).

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